

Tyler Pace

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EDUCATION

BS: Applied and Computational Mathematics, Emphasis in Mathematical Biology

April 2026

Brigham Young University

Provo, UT

- GPA 3.6
- Relevant Coursework: Mathematical Analysis, Algorithm Design and Optimization, Advanced Linear Algebra, Modeling with Data and Uncertainty, Modeling with Dynamics and Control, Mathematical Biology, Molecular Biology, Genetics

SKILLS

Languages and Tools – Python, C++, SQL, LaTeX

Libraries – NumPy, Pandas, SciPy, Scikit-Learn, iPyParallel, PyTorch, Matplotlib, Seaborn, NetworkX

Machine Learning – *Supervised*: Random Forests, Linear and Logistic Regression, Naïve Bayes, GDA, LDA, SVM, Kernel Methods | *Unsupervised*: GMM, K-Means, HMM | *Dimensionality Reduction*: PCA, t-SNE, UMAP, NMF | *Deep Learning*: PyTorch Model Development and Validation, Hyperparameter Tuning.

Statistical Modeling – Probability Theory, Network Theory, Causal Inference Theory, Bayesian Inference, Time Series (ARIMA, Kalman filters), Sampling (Importance, Rejection, MCMC)

Mathematical Techniques – Numerical Methods, Dynamic Optimization, Optimal Control Theory, Advanced Linear Algebra, Signal Processing (Fourier Analysis, Wavelets)

PROJECTS

colony-opt (github.com/tpace31/colony-opt) | *Python, Optimal Control Theory, ODE's, SciPy*

April 2026

- Modeled dynamic resource allocation and reproduction strategies between rival ant colonies

pharma-state (github.com/tpace31/pharma-state) | *Python, Kalman Filters, Random Forest, Pharmacokinetics*

April 2026

- Engineered a machine learning pipeline using a Kalman filter and random forest to extract and predict personalized pharmacological parameters

glymphatic-ode (github.com/tpace31/glymphatic-ode) | *Compartmental Modeling, Lyapunov Stability*

December 2025

- Developed a compartmental ODE model to quantify toxic metabolite buildup from sleep deprivation

taxi-predict (github.com/tpace31/taxi-predict) | *Python, Random Forest Regressor, Feature Engineering*

December 2025

- Optimized a random forest regressor using target encoding on NYC taxi records to predict route durations

WORK EXPERIENCE

Resident Assistant

May 2024 – April 2026

Brigham Young University, Residence Life, Helaman Halls

Provo, UT

- Managed 24/7 crisis response and community programming for 40+ first-year students, earning the department's Employee of the Year award (2025) for exceptional leadership

Repair Specialist

May 2025 – August 2025

Safelite AutoGlass

Provo, UT

- Independently managed a high-volume service route handling high-value assets while adapting to challenging conditions and delivering excellent face-to-face customer service

Research Assistant (Global Value Chain)

May 2023 – August 2024

Brigham Young University, Mathematics Department

Provo, UT

- Managed and streamlined a data pipeline encoding daily information from 180,000+ firms, creating networks with 415,000+ edges for analysis
- Developed statistical tools and algorithms in Python to ensure the accuracy of data gathered and enable dynamic community detection for insights
- Collaborated efforts and organized findings with team, presenting at the Conference on Complex Systems in Brazil, October 2023, "Exploring the Dynamic, Global Value Chain"

ADDITIONAL

Languages: Spanish (fluent), German (beginner)

Interests: Travel and exploration, automotive repair, storytelling and comedy, omnivorous reading